1. Aim

To design a responsive front-end prototype of an online train reservation system with train search, seat availability, booking with PNR generation, PNR status, cancellation with refund rules and admin management.

2. Technologies Used

Component	Technology
Front end	HTML5, CSS3 (responsive layout), JavaScript
Data	Sample data held in JavaScript variables (mock-up, no back end)
Editor	Notepad / Visual Studio Code
Browser	Google Chrome or Microsoft Edge

3. Modules and Features

Passenger

- Login with valid credentials
- Search trains by origin, destination, date and class; view train list, seat availability and via stations
- Enter passenger details and pay online or offline; ticket and PNR are generated
- Check PNR status (Confirmed, RAC, Waiting)
- Cancel ticket by PNR; refund calculated from refund rules
- View booking history and print ticket

Admin

- Maintain train details, class-wise fares and seats (weekend date-wise fare +10%)
- View payment details of every ticket
- Pass the refundable amount for cancelled tickets

Refund rules

- More than 2 days before journey: 80%
- 1 to 2 days before journey: 50%
- On the day of journey: 25%

4. Algorithm

1. Passenger searches by From, To, Date and Class; the fare is base fare (x1.1 on Saturday and Sunday).
2. Availability = seats of the class - confirmed passengers on that date.
3. On booking: if availability >= passengers then Confirmed, else RAC (limited quota), else Waiting.
4. A 10-digit PNR is generated, the payment mode (Online / Offline) and amount are recorded for the admin.
5. Cancellation: refund % is selected from the days left to journey; a refund request is created.
6. Admin passes the refund and the ticket status becomes Refunded.

5. Source Code

exp4_train.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>RailBook - Train Reservation</title>
<style>
:root{--c:#1a5276}
*{box-sizing:border-box}
body{margin:0;font-family:"Segoe UI",Arial,sans-serif;background:#f1f5f9;color:#222}
header{background:var(--c);color:#fff;padding:10px 16px;display:flex;flex-wrap:wrap;gap:10px;align-
items:center;justify-content:space-between}
.brand{font-size:1.25rem;font-weight:600}
nav{display:flex;flex-wrap:wrap;gap:4px}
nav a{color:#fff;opacity:.92;padding:6px 10px;border-radius:6px;font-size:.9rem;cursor:pointer}
nav a:hover{background:rgba(255,255,255,.2)}
main{max-width:1000px;margin:20px auto;padding:0 16px}
h1,h2{color:var(--c);margin-top:0}
.card{background:#fff;border-radius:8px;padding:18px;box-shadow:0 1px 4px rgba(0,0,0,.12);margin-
bottom:16px}
.narrow{max-width:440px;margin:24px auto}
label{display:block;margin:10px 0 4px;font-weight:600;font-size:.9rem}
input,select{width:100%;padding:8px;border:1px solid #bbb;border-radius:5px;font:inherit}
button{background:var(--c);color:#fff;border:0;padding:8px 14px;border-
radius:5px;cursor:pointer;font:inherit}
button.red{background:#b8423a}button.grey{background:#777}button.s{padding:4px 9px;font-size:.82rem}
button:disabled{background:#bbb;cursor:not-allowed}
.row{display:flex;flex-wrap:wrap;gap:10px;align-items:end}.row>*{flex:1;min-width:120px}.row
button{flex:0 0 auto}
table{width:100%;border-collapse:collapse;background:#fff}
th,td{padding:8px 10px;border-bottom:1px solid #e5e5e5;text-align:left;font-size:.9rem}
th{background:#d6e6f2}
.tw{overflow-x:auto}
.ok{color:#1d7a35;font-weight:600}.bad{color:#b8423a;font-weight:600}.warn{color:#b9770e;font-
weight:600}.muted{color:#666;font-size:.88rem}
.msg{padding:10px 14px;border-radius:6px;margin-
bottom:12px}.msg.s{background:#def3e2;color:#1d5a2a}.msg.e{background:#fbe0de;color:#8a2822}
.ticket{border:2px dashed var(--c);padding:16px;border-radius:8px;background:#f7fbff}
.pnr{font-size:1.8rem;letter-spacing:2px;color:var(--c);font-weight:700}
.stat{display:inline-block;background:#fff;border-left:5px solid var(--c);padding:12px 18px;margin:0
10px 10px 0;box-shadow:0 1px 4px rgba(0,0,0,.12)}
.stat b{display:block;font-size:1.5rem}
</style>
</head>
<body>
<header><div class="brand">&#128646; RailBook</div><nav id="nav"></nav></header>
<main>
<div id="msg"></div>

<section class="page" id="login"><div class="card narrow"><h1>Login</h1>
  <label>Username</label><input id="l_u"><label>Password</label><input id="l_p" type="password">
  <label>Login as</label><select id="l_r"><option value="user">Passenger</option><option
value="admin">Admin</option></select>
  <p><button onclick="doLogin()">Login</button></p>
  <p class="muted">Demo: any credentials work. New user? <a href="#" onclick="show('register');return
false">Register</a></p></div></section>
<section class="page" id="register"><div class="card narrow"><h1>Register</h1>
  <label>Full name</label><input id="r_n"><label>Email</label><input id="r_e"
type="email"><label>Username</label><input id="r_u"><label>Password</label><input id="r_p"
type="password">
  <p><button onclick="doRegister()">Register</button></p></div></section>

<!-- ========== PASSENGER ========== -->
<section class="page" id="search">
  <div class="card"><h1>Search Trains</h1><div class="row">
    <div><label>From station</label><select id="s_f"></select></div><div><label>To
station</label><select id="s_t"></select></div>
    <div><label>Date of journey</label><input type="date" id="s_d"></div>
    <div><label>Class</label><select id="s_c"></select></div><button onclick="searchTrains()">Find
Trains</button></div></div>
  <div id="results"></div></section>

<section class="page" id="book"><div class="card"><h1>Passenger Details</h1><p id="bk_info"></p>
  <div id="bk_rows"></div><p><button class="grey s" onclick="addPax()">+ Add passenger</button></p>
```

```
  <h2>Payment</h2><div class="row"><div><label>Payment mode</label><select id="bk_pm"
onchange="payMode()"><option value="Online">Online (card / UPI)</option><option
value="Offline">Offline (pay at counter)</option></select></div>
  <div id="card_box"><label>Card / UPI id</label><input id="bk_card" placeholder="1234 5678 9012 3456
or name@upi"></div></div>
  <p><b id="bk_total"></b></p><p><button onclick="payBook()">Pay &amp; Generate Ticket</button>
<button class="grey" onclick="show('search')">Back</button></p></div></section>

<section class="page" id="ticket"><div class="card narrow"><h1>Ticket Generated</h1><div
class="ticket" id="tk"></div>
  <p class="muted">Payment details have been sent to the system admin. Confirmation sent to your
email.</p><p><button onclick="window.print()">Print Ticket</button> <button class="grey"
onclick="show('mybookings')">My Bookings</button></p></div></section>

<section class="page" id="pnr"><div class="card narrow"><h1>PNR Status</h1><label>Enter 10-digit
PNR</label><input id="pn_n" maxlength="10">
  <p><button onclick="checkPnr()">Check Status</button></p><div id="pn_res"></div></div></section>

<section class="page" id="cancel"><div class="card narrow"><h1>Cancel Ticket</h1>
  <p class="muted">Refund rules: more than 2 days before journey = 80% refund; 1-2 days = 50%; on the
day = 25%.</p>
  <label>PNR number</label><input id="cn_n" maxlength="10"><p><button onclick="findCancel()">Find
Ticket</button></p><div id="cn_res"></div></div></section>

<section class="page" id="mybookings"><h1>My Booking History</h1><div class="tw"
id="mb"></div></section>

<!-- ========== ADMIN ========== -->
<section class="page" id="a_trains"><h1>Manage Trains, Fares &amp; Seats</h1>
  <p class="muted">Fares can be set per class. Weekend journeys (Sat/Sun) cost 10% extra (date-wise
fare).</p><div class="tw" id="at"></div></section>
<section class="page" id="a_pay"><h1>Payment Details</h1><div id="ap_stats"></div><div class="tw"
id="ap"></div></section>
<section class="page" id="a_refund"><h1>Refund Requests</h1><div class="tw" id="ar"></div></section>
</main>

<script>
function $(i){return document.getElementById(i)}
function pad(n){return (n<10?'0':'')+n}
function fmt(d){return d.getFullYear()+'-'+pad(d.getMonth()+1)+'-'+pad(d.getDate())}
function addDays(n){var d=new Date();d.setDate(d.getDate()+n);return fmt(d)}
function days(a,b){var x=a.split('-'),y=b.split('-');return Math.round((Date.UTC(+y[0],+y[1]-1,+y[2])-
Date.UTC(+x[0],+x[1]-1,+x[2]))/864e5)}
function msg(t,ok){$('msg').innerHTML='<div class="msg
'+(ok===false?'e':'s')+'">'+t+'</div>';window.scrollTo(0,0)}
function
show(id){document.querySelectorAll('.page').forEach(function(p){p.hidden=p.id!==id});$('msg').innerHTM
L='';window.scrollTo(0,0);if(window['on_'+id])window['on_'+id]();drawNav()}
function esc(s){return String(s).replace(/[&<>"]/g,function(c){return
{'&':'&amp;','<':'&lt;','>':'&gt;','"':'&quot;'}[c]})}

// ---------------------------------------------- data
var CLASSES=['SL','3A','2A','1A'], RAC_LIMIT=2;
var trains=[
 {no:'12601',name:'Chennai Mail',from:'Chennai',to:'Bangalore',via:'Katpadi,
Jolarpet',dep:'06:00',arr:'11:30',fare:{SL:250,'3A':650,'2A':900,'1A':1500},seats:{SL:40,'3A':20,'2A':
10,'1A':4}},
 {no:'12027',name:'Shatabdi
Express',from:'Chennai',to:'Bangalore',via:'Katpadi',dep:'15:30',arr:'20:30',fare:{SL:300,'3A':720,'2A
':980,'1A':1650},seats:{SL:30,'3A':15,'2A':8,'1A':2}},
 {no:'12433',name:'Rajdhani Express',from:'Chennai',to:'Delhi',via:'Vijayawada, Nagpur,
Agra',dep:'22:00',arr:'06:00',fare:{SL:900,'3A':1900,'2A':2600,'1A':4300},seats:{SL:50,'3A':30,'2A':14
,'1A':6}},
 {no:'12608',name:'Lalbagh Express',from:'Bangalore',to:'Chennai',via:'Jolarpet,
Katpadi',dep:'06:30',arr:'11:50',fare:{SL:240,'3A':640,'2A':890,'1A':1480},seats:{SL:40,'3A':20,'2A':1
0,'1A':4}}];
var users=[], tickets=[], refunds=[], nextPnr=2410001000, me=null, cur=null, curClass='SL',
curDate='';
var REFUND_RULES=[{min:3,pct:80},{min:1,pct:50},{min:0,pct:25}];

function drawNav(){
  var l;
  if(!me)l=[['login','Login'],['register','Register']];
  else if(me.role==='user')l=[['search','Search Trains'],['pnr','PNR Status'],['cancel','Cancel
Ticket'],['mybookings','My Bookings'],['out','Logout']];
  else l=[['a_trains','Trains &amp; Fares'],['a_pay','Payments'],['a_refund','Refunds'],['pnr','PNR
Status'],['out','Logout']];
```

```
  $('nav').innerHTML=l.map(function(x){return '<a
onclick="'+(x[0]==='out'?'logout()':"show('"+x[0]+"')")+'">'+x[1]+'</a>'}).join('');
}
function logout(){me=null;show('login');msg('Logged out.')}
function doLogin(){
  if(!$('l_u').value.trim()||!$('l_p').value)return msg('Enter valid credentials.',false);

me={u:$('l_u').value.trim(),role:$('l_r').value};show(me.role==='admin'?'a_trains':'search');msg('Welc
ome, '+esc(me.u)+'!');
}
function doRegister(){
  if(!$('r_n').value.trim()||!$('r_u').value.trim()||!$('r_p').value)return msg('Fill in all required
fields.',false);
  users.push($('r_u').value.trim());show('login');msg('Registration successful. Please log in.');
}

// ------------------------------------------------ search & seat availability
function fare(t,c,date){var d=new Date(date+'T00:00:00').getDay();return
Math.round(t.fare[c]*((d===0||d===6)?1.1:1))}
function used(t,c,date){return tickets.filter(function(k){return
k.train===t.no&&k.cls===c&&k.date===date&&k.status!=='Cancelled'&&k.status!=='Waiting'&&k.status!=='RA
C'}).reduce(function(s,k){return s+k.pax.length},0)}
function avail(t,c,date){return Math.max(0,t.seats[c]-used(t,c,date))}
function on_search(){
  if(!$('s_f').options.length){
    var
st=[];trains.forEach(function(t){[t.from,t.to].forEach(function(p){if(st.indexOf(p)<0)st.push(p)})});s
t.sort();
    var o=st.map(function(p){return
'<option>'+p+'</option>'}).join('');$('s_f').innerHTML=o;$('s_t').innerHTML=o;$('s_t').selectedIndex=M
ath.min(1,st.length-1);
    $('s_c').innerHTML=CLASSES.map(function(c){return
'<option>'+c+'</option>'}).join('');$('s_d').value=addDays(5);
  }
}
function searchTrains(){
  var f=$('s_f').value,t=$('s_t').value,c=$('s_c').value,d=$('s_d').value;
  if(!d||d<fmt(new Date()))return msg('Choose a valid journey date.',false);
  var r=trains.filter(function(x){return x.from===f&&x.to===t});
  $('results').innerHTML=r.length?r.map(function(x){var a=avail(x,c,d);
    return '<div class="card"><div class="row"><div><h2>'+esc(x.name)+' ('+x.no+')</h2><span
class="muted">'+x.from+' '+x.dep+' &rarr; '+x.to+' '+x.arr+'<br>Via: '+esc(x.via)+'</span></div>'+
    '<div><b>Class '+c+': Rs. '+fare(x,c,d)+'</b><br>'+(a>0?'<span class="ok">Available
'+a+'</span>':'<span class="warn">Waiting list</span>')+'</div>'+
    '<button
onclick="startBook(\''+x.no+'\',\''+c+'\',\''+d+'\')">Book</button></div></div>'}).join(''):'<div
class="card">No trains on this route.</div>';
}

// ------------------------------------------------ booking
function startBook(no,c,d){cur=trains.find(function(t){return
t.no===no});curClass=c;curDate=d;show('book')}
function on_book(){
  $('bk_info').innerHTML='<b>'+esc(cur.name)+' ('+cur.no+')</b><br>'+cur.from+' &rarr; '+cur.to+' on
'+curDate+', class '+curClass+' - Rs. '+fare(cur,curClass,curDate)+' per passenger';
  $('bk_rows').innerHTML='';addPax();payMode();
}
function addPax(){
  if($('bk_rows').children.length>=6)return msg('Maximum 6 passengers per booking.',false);
  var i=$('bk_rows').children.length+1,d=document.createElement('div');d.className='row';
  d.innerHTML='<div><label>Passenger '+i+' name</label><input
class="pn"></div><div><label>Age</label><input class="pa" type="number" min="1"
max="120"></div><div><label>Gender</label><select
class="pg"><option>Male</option><option>Female</option><option>Other</option></select></div>';
  $('bk_rows').appendChild(d);$('bk_total').textContent='Total fare: Rs.
'+($('bk_rows').children.length*fare(cur,curClass,curDate));
}
function payMode(){$('card_box').hidden=$('bk_pm').value!=='Online'}
function payBook(){
  var
names=[].slice.call(document.querySelectorAll('.pn')),ages=[].slice.call(document.querySelectorAll('.p
a')),gens=[].slice.call(document.querySelectorAll('.pg')),pax=[];
  for(var i=0;i<names.length;i++){if(!names[i].value.trim()||!(ages[i].value>0))return msg('Enter name
and age for every
passenger.',false);pax.push({n:names[i].value.trim(),a:+ages[i].value,g:gens[i].value})}
  var pm=$('bk_pm').value;if(pm==='Online'&&!$('bk_card').value.trim())return msg('Enter card / UPI
details.',false);
  var a=avail(cur,curClass,curDate),status;
  if(a>=pax.length)status='Confirmed';
```

```
  else{var rac=tickets.filter(function(k){return
k.train===cur.no&&k.cls===curClass&&k.date===curDate&&k.status==='RAC'}).reduce(function(s,k){return
s+k.pax.length},0);status=(rac+pax.length<=RAC_LIMIT*4)?'RAC':'Waiting'}
  var amt=pax.length*fare(cur,curClass,curDate),pnr=String(nextPnr++);

tickets.push({pnr:pnr,train:cur.no,tname:cur.name,from:cur.from,to:cur.to,date:curDate,cls:curClass,pa
x:pax,amount:amt,mode:pm,status:status,by:me.u,booked:fmt(new Date())});
  show('ticket');
  $('tk').innerHTML='<div class="pnr">PNR '+pnr+'</div><b>'+esc(cur.name)+'
('+cur.no+')</b><br>'+cur.from+' &rarr; '+cur.to+'<br>Journey: '+curDate+' &middot; Class
'+curClass+'<br>Passengers: '+pax.map(function(p){return esc(p.n)+' ('+p.a+')'}).join(', ')+
    '<br>Status: <b>'+status+'</b><br>Paid ('+pm+'): <b>Rs. '+amt+'</b>';
}

// ---------------------------------------------- PNR + history + cancel
function statusCls(s){return s==='Confirmed'?'ok':(s==='Cancelled'||s==='Refunded'?'bad':'warn')}
function checkPnr(){
  var k=tickets.find(function(x){return x.pnr===$('pn_n').value.trim()});
  $('pn_res').innerHTML=k?'<div class="ticket"><b>'+esc(k.tname)+'</b><br>'+k.from+' &rarr; '+k.to+',
'+k.date+' ('+k.cls+')<br>Passengers: '+k.pax.length+'<br>Status: <b
class="'+statusCls(k.status)+'">'+k.status+'</b></div>':'<p class="bad">PNR not found.</p>';
}
function on_mybookings(){
  var r=tickets.filter(function(k){return k.by===me.u}).map(function(k){
    return '<tr><td>'+k.pnr+'</td><td>'+esc(k.tname)+'</td><td>'+k.from+' &rarr;
'+k.to+'</td><td>'+k.date+'</td><td>'+k.cls+'</td><td>'+k.pax.length+'</td><td>Rs.
'+k.amount+'</td><td class="'+statusCls(k.status)+'">'+k.status+'</td></tr>'}).join('')||'<tr><td
colspan="8">No bookings yet.</td></tr>';

$('mb').innerHTML='<table><tr><th>PNR</th><th>Train</th><th>Route</th><th>Date</th><th>Class</th><th>P
ax</th><th>Fare</th><th>Status</th></tr>'+r+'</table>';
}
function refundPct(date){var d=days(fmt(new Date()),date);for(var
i=0;i<REFUND_RULES.length;i++)if(d>=REFUND_RULES[i].min)return REFUND_RULES[i].pct;return 0}
function findCancel(){
  var k=tickets.find(function(x){return x.pnr===$('cn_n').value.trim()&&x.by===me.u});
  if(!k)return $('cn_res').innerHTML='<p class="bad">No ticket with this PNR in your account.</p>';
  if(k.status==='Cancelled'||k.status==='Refunded')return $('cn_res').innerHTML='<p class="bad">This
ticket is already cancelled.</p>';
  var p=refundPct(k.date);
  $('cn_res').innerHTML='<div class="ticket"><b>'+esc(k.tname)+'</b> on '+k.date+'<br>Fare paid: Rs.
'+k.amount+'<br>Refundable: '+p+'% = <b>Rs. '+Math.round(k.amount*p/100)+'</b></div><p><button
class="red" onclick="doCancel(\''+k.pnr+'\')">Confirm Cancel Request</button></p>';
}
function doCancel(pnr){
  var k=tickets.find(function(x){return x.pnr===pnr}),p=refundPct(k.date);

k.status='Cancelled';refunds.push({pnr:pnr,by:k.by,amount:Math.round(k.amount*p/100),pct:p,state:'Pend
ing'});
  $('cn_res').innerHTML='';$('cn_n').value='';msg('Ticket cancelled. Refund of Rs.
'+refunds[refunds.length-1].amount+' will be released after admin approval.');
}

// ---------------------------------------------- admin
function on_a_trains(){

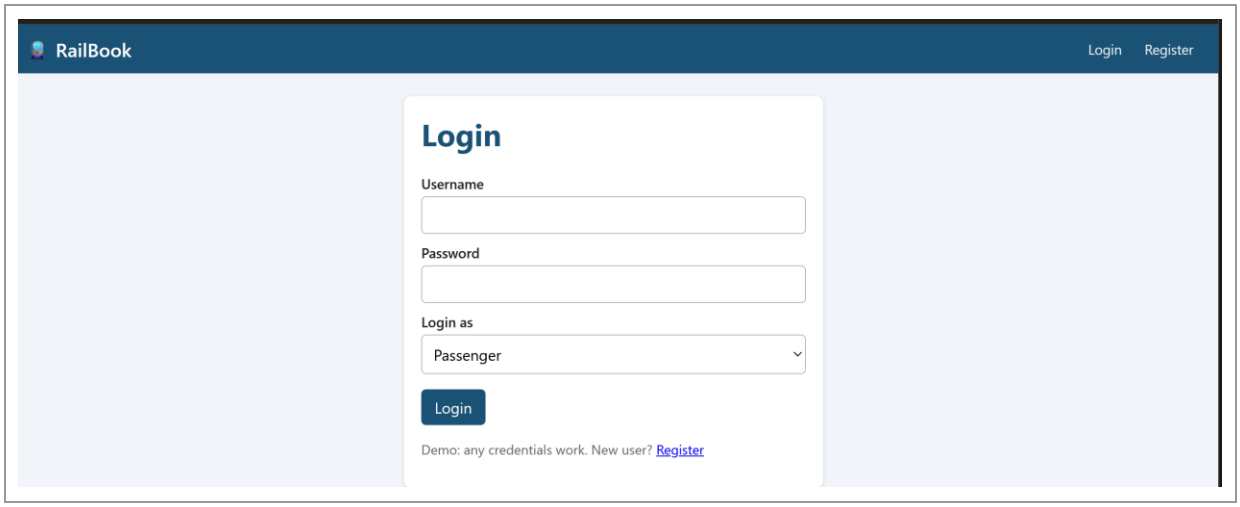
$('at').innerHTML='<table><tr><th>No.</th><th>Train</th><th>Route</th>'+CLASSES.map(function(c){return
'<th>'+c+' fare / seats</th>'}).join('')+'<th></th></tr>'+trains.map(function(t,i){
    return '<tr><td>'+t.no+'</td><td>'+esc(t.name)+'</td><td>'+t.from+' &rarr;
'+t.to+'</td>'+CLASSES.map(function(c){return '<td><input id="f'+i+c+'" type="number"
value="'+t.fare[c]+'" style="width:70px"> / <input id="s'+i+c+'" type="number" value="'+t.seats[c]+'"
style="width:60px"></td>'}).join('')+'<td><button class="s"
onclick="saveTrain('+i+')">Save</button></td></tr>'}).join('')+'</table>';
}
function
saveTrain(i){CLASSES.forEach(function(c){trains[i].fare[c]=+$('f'+i+c).value;trains[i].seats[c]=+$('s'
+i+c).value});msg('Train '+trains[i].no+' updated.')}
function on_a_pay(){
  var tot=tickets.reduce(function(s,k){return s+k.amount},0);
  $('ap_stats').innerHTML='<div class="stat"><b>'+tickets.length+'</b>Tickets booked</div><div
class="stat"><b>Rs. '+tot+'</b>Total collected</div>';
  $('ap').innerHTML='<table><tr><th>PNR</th><th>Passenger
account</th><th>Train</th><th>Date</th><th>Mode</th><th>Amount</th><th>Status</th></tr>'+(tickets.map(
function(k){
    return
'<tr><td>'+k.pnr+'</td><td>'+esc(k.by)+'</td><td>'+esc(k.tname)+'</td><td>'+k.date+'</td><td>'+k.mode+
'</td><td>Rs. '+k.amount+'</td><td
class="'+statusCls(k.status)+'">'+k.status+'</td></tr>'}).join('')||'<tr><td colspan="7">No payments
yet.</td></tr>')+'</table>';
}
```

```
function on_a_refund(){
  $('ar').innerHTML='<table><tr><th>PNR</th><th>Passenger</th><th>Refund %</th><th>Refund
amount</th><th>State</th><th></th></tr>'+(refunds.map(function(r,i){
    return '<tr><td>'+r.pnr+'</td><td>'+esc(r.by)+'</td><td>'+r.pct+'%</td><td>Rs.
'+r.amount+'</td><td class="'+(r.state==='Pending'?'warn':'ok')+'">'+r.state+'</td><td><button
class="s" '+(r.state==='Pending'?'':'disabled')+' onclick="payRefund('+i+')">Pass
Refund</button></td></tr>'}).join('')||'<tr><td colspan="6">No refund
requests.</td></tr>')+'</table>';
}
function payRefund(i){refunds[i].state='Refunded';var k=tickets.find(function(x){return
x.pnr===refunds[i].pnr});if(k)k.status='Refunded';on_a_refund();msg('Refund of Rs.
'+refunds[i].amount+' passed to the passenger.')}

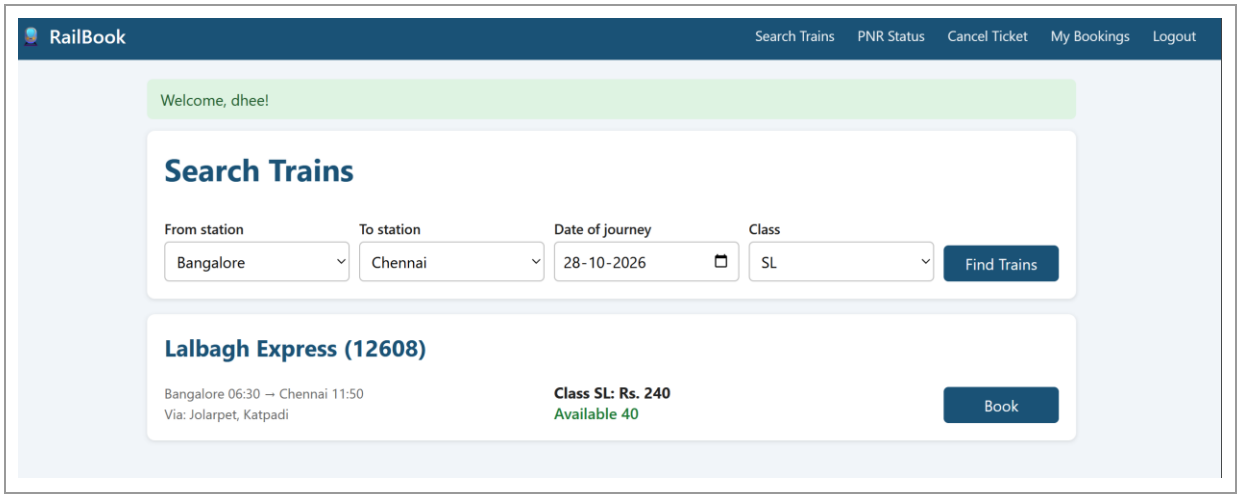
show('login');
</script>
</body>
</html>
```

7. Output

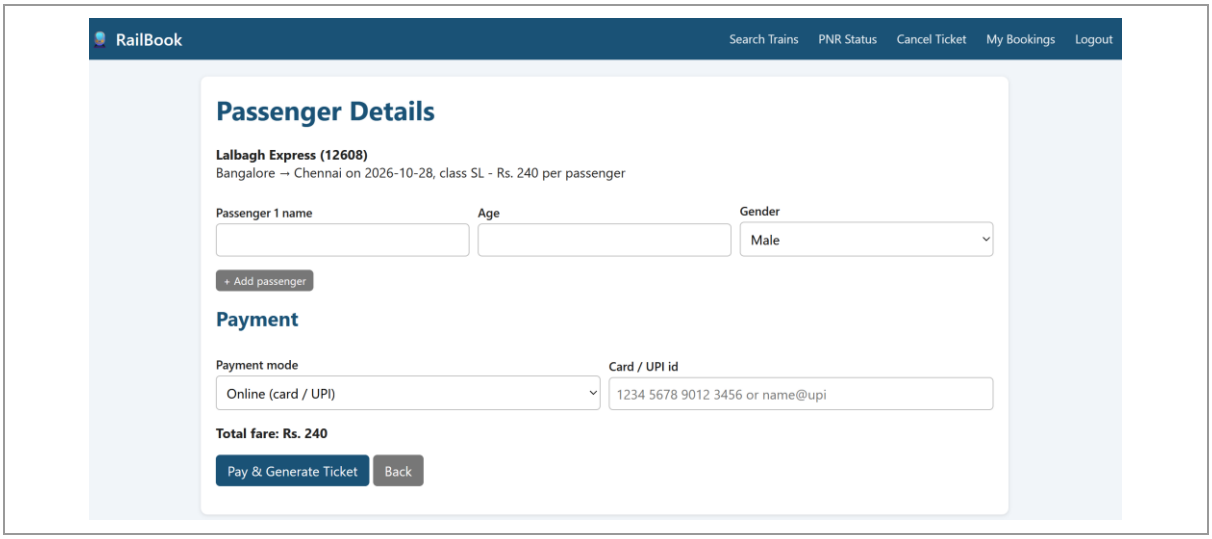
Output 1: Login page



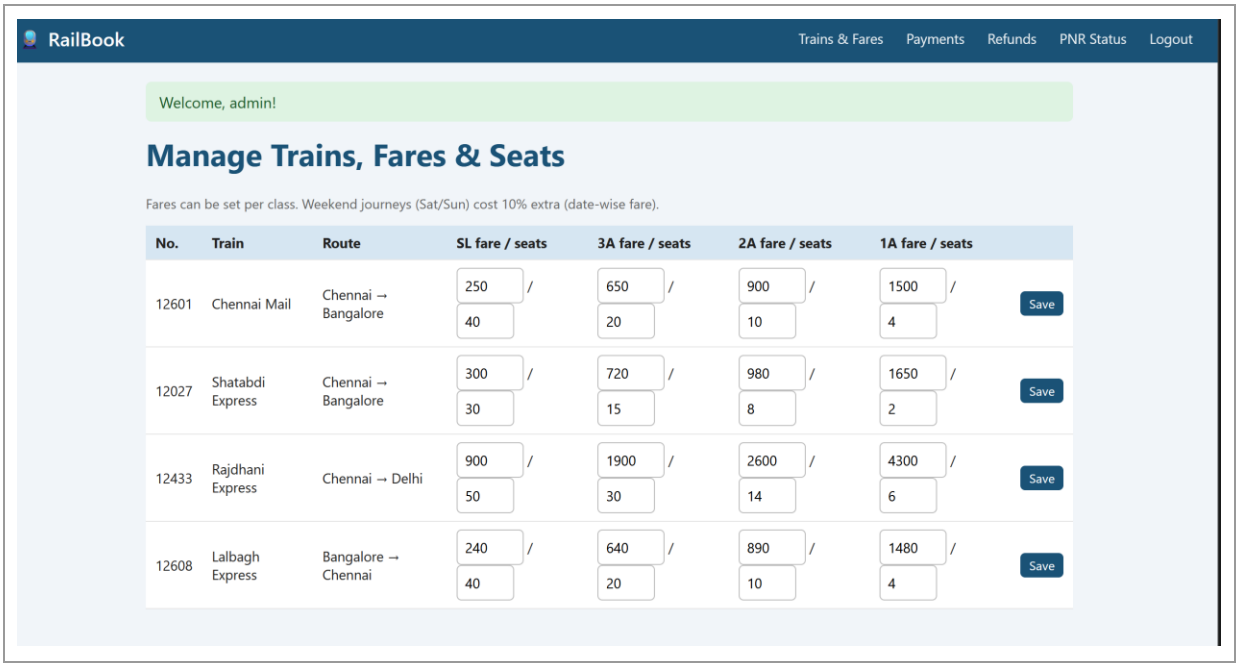
Output 2: Search trains



Output 3: Passenger details and payment



Output 4: Admin: trains and fares



8. Result

The online train reservation system prototype was designed and tested successfully; all the listed pages and features work in the browser.